

Econometric Theory and Machine Learning

Group Assignment (30 marks)

Kindergarten Environment & Achievements

Background

Australia is in the midst of major early-childhood reforms aimed at improving affordability, access and quality. In that context, robust evidence on how classroom conditions shape learning is directly policy relevant. A landmark source of such evidence is Tennessee's *Project STAR* experiment, in which more than 11,000 students in kindergarten through third grade (and their teachers) were randomly assigned *within schools* to one of three settings: *small* classes (13–17 students), *regular* classes (22–25 students), or *regular classes with a teacher's aide*. Because assignment occurred within schools, differences in outcomes can credibly be attributed to class size rather than school or neighbourhood characteristics.

Project STAR shows that students placed in smaller classes achieved higher test scores in the early years; remarkably, the same cohort experienced substantial gains in adulthood. Chetty *et al.* (2011) link the original STAR records to U.S. tax data and find that kindergarten classroom quality predicts higher earnings, greater college attendance, and improved neighbourhood outcomes in later life (the paper can be found here: https://www.hks.harvard.edu/sites/default/files/centers/taubman/files/kindergarten_final.pdf and I strongly recommend reading it). While the institutional setting may differ from Australia, the mechanism—more individual attention in smaller classes—speaks directly to current debates on educator-child ratios and classroom quality.

These findings motivate the empirical work in this assignment: using a cross-section of kindergarten students, you will build transparent, well-validated *predictive* models of achievement and *interpret* the most important drivers of performance.

Data

You will receive link to `sample_kinder_data.csv` data file containing the following variables.

Variables

Name	Type	Description
<code>score_read</code>	numeric	Student reading score.
<code>score_math</code>	numeric	Student math score.
<code>gender</code>	categorical	Student gender: male or female .
<code>ethnicity</code>	categorical	cauc =Caucasian/White; afam =African American/Black; asian =Asian; hispanic =Hispanic/Latino; amindian =American Indian or Alaska Native; other =other/mixed.
<code>birth</code>	yearqtr	Student birth quarter (year-quarter; original coding).
<code>lunch</code>	categorical	Free-lunch eligibility (yes/no), proxy for SES.
<code>class_type</code>	categorical	Class type in kindergarten: regular , small , regular+aide .
<code>school</code>	categorical	School type: inner-city , suburban , rural , urban .
<code>degree</code>	categorical	Teacher highest degree: bachelor , master , specialist , master+ .
<code>ladder</code>	categorical	Teacher career ladder level: probation < apprentice < level1 < level2 < level3 .
<code>experience</code>	numeric	Teacher years of experience.
<code>t_ethnicity</code>	categorical	Teacher ethnicity, see the definitions above.
<code>schooldistrict_id</code>	categorical	School district identifier.
<code>school_id</code>	categorical	School identifier.

Task objective

Using `sample_kinder_data.csv`, build models to predict `score_read` and `score_math`, either separately or within a joint specification, from the available covariates (indeed you may use variable selection) and *interpret* the most important drivers of performance.

After you submit, out-of-sample performance of one model you select will be assessed on `out_kinder_data.csv` (withheld), which has the same variables as the sample data.

Let $y_{\text{read},i}$, $y_{\text{math},i}$ be the true scores and $\hat{y}_{\text{read},i}$, $\hat{y}_{\text{math},i}$ your predictions:

$$\text{MSE}_o = \frac{1}{2n_o} \sum_{i=1}^{n_o} [(\hat{y}_{\text{read},i} - y_{\text{read},i})^2 + (\hat{y}_{\text{math},i} - y_{\text{math},i})^2].$$

3 groups/individuals that receive the **smallest** MSE_o will receive 2 extra **bonus marks**.

Specifically, you must:

1. Conduct exploratory data analysis (visualisation incl. bivariate plots, missing data).
2. Fit at least **five** different models including ordinary least squares, regularised linear models, non-linear (machine learning) models (e.g., random forest or neural net).
3. Select a final model using appropriate model selection criteria.
4. Explain drivers for each **score** and conduct appropriate statistical testing or analogous diagnostics.

Software

The group leader must create a Colab notebook, name it `implementation-xxx.ipynb`, where `xxx` is your group name as in Moodle (if you work alone, your student ID) and share it with all group members. In Colab, click **Share** and set General access to Anyone with the link and set the role to **Editor**. Then, share the link with all group members. This way everyone is able to collaborate on the assignment.

Required cells in your notebook: Include this code in your notebook. For now use the same url for out-of-sample dataset to make sure your code works and produces `mse_out`. When you submit, the `out_url` link will be changed to the actual out-of-sample dataset by markers.

```
import pandas as pd

# load data
sample_url=
"https://drive.google.com/uc?export=download&id=1ekCjJOTjGD00ASerwVnwhJuPD0hFfCEF"

out_url=sample_url

db_sample = pd.read_csv(sample_url)
db_out = pd.read_csv(out_url)

# your code...

print(mse_out)
```

Deliverables

1. **Report** (max length 20 pages, **20 marks**)
 - **Summary of the task and main findings:** (1 page or less)

- **Exploratory data analysis (EDA):** visualisation, correlations, and initial insights.
- **Models:** brief description of each model (out of 5 models) and justification for the selected model.
- **Drivers of performance:** identify the most important features/covariates that drive performance (you may use linear models and standard tests), discuss any omitted variables and whether they may bias your results (note that random assignment helps, why?).
- **Discussion:** model interpretability, caveats.
- **References:** including referencing AI and statement on AI use. See <https://www.unsw.edu.au/student/managing-your-studies/academic-skills-support/toolkit/referencing/ai-referencing>
- Note: based on the UNSW AI policy, this assignment allows “Assistance with attribution”.
- Include relevant graphs and tables within the relevant sections, but not the Python code.
- I recommend shareable Google Docs (or MS Word) for the report (it is easy to copy/paste figures from Google Colab) but if you are after more of a challenge, you may use LaTeX. Transform to pdf for submission.

2. Implementation (7 marks)

Reproducible code that reads data, produces EDA, estimates models and selects one, performs necessary inference, computes and prints the out-of-sample MSE.

3. Reflections (one page or less, 3 marks)

If you are working in the group, include brief **meeting notes** for each (online) meeting (date, time, who was present/what was discussed) and any possible **issues** you faced working in the group, substantial differences in contributions. Any substantial underperformance in a group may lead to a lower mark for the group member.

If you work alone, you need to explain why you prefer working alone, discuss the assignment at least with one person (may be another group member), indicate the date/time of this meeting; and reflect on most challenging part of the assignment.

Submission

Only one group member needs to submit the assignment via Moodle site. You need to coordinate who will do the submission, if you work in a group. The submission will be automatically populated to all other group members. You need to submit at least 3 files: `report-xxx.pdf`, `implementation-xxx.ipynb`, `reflect-xxx.pdf`, where `xxx` is your group name as in Moodle; if you work alone, your student ID. In addition, submit the link to your Google Colab notebook, `implementation-xxx.ipynb`.

Deadline: Sun, 16 Nov, 22:00